

Calculus I (MAT 1001), L02

Syllabus

Fall, 2025, CUHK-SZ

Instructor:	刘杨 Prof. Yang LIU
Office:	Zhiren 404
Office hours:	Friday at 1:30-2:30 pm or by appointment
Email:	yangliu16@cuhk.edu.cn
Lecture Time and Location	Mon & Wed 15:30 -17:20, J_E205

Teaching Assistants:	林源盛 LIN Yuansheng	金晟远 JIN Shengyuan	薛洁 XUE Jie
Tutorial Time and Location:	Tue 18:00-19:30, 19:30-21:00, TA316	Tue 18:00-19:30, 19:30-21:00, TA308	Tue 19:00-20:30, TB304
Office hours:	Tue 21:00-22:00, RB402	Tue 21:00-22:00, Zhiren321	Tue 20:30-21:30, RB514
Emails:	223010100@link.cuhk.edu.cn	120090073@link.cuhk.edu.cn	222013005@link.cuhk.edu.cn
Class:	T09 and T10	T06 and T07	T08

Text: *Thomas' Calculus (13th Edition in SI Units)*, George B. Thomas Jr., Maurice D. Weir, Joel R. Hass. Publisher: Pearson.

Prerequisites: High School Algebra, Trigonometry, Geometry (In particular, Chapter 1 of the above textbook)

Reference: *Calculus: Early Transcendentals (8th Edition)*, James Stewart.
Publisher: Brooks Cole.

Course Description: This course covers one-variable calculus; it emphasizes intuitive and conceptual understanding of the theory of Calculus, computation skills, and nurtures the mentality and the ability to use Calculus to solve problems in other scientific disciplines.

Course Policies:

- The **semester grade** will be given according to performance in **homework (10%), weekly quizzes (25%), mid-term (25%) and the final exam (40%).**
- In addition to lectures, each student is assigned with a tutorial each week. Tutorials will be conducted by teaching assistants (TAs). **No tutorial in the first week (Sept 1-5).**
- **Homework** will be assigned and graded weekly. Each homework assignment covers the sections taught in the lecture class in the same week and **must be handed in** in the **tutorial** class of the following week. No late homework will be accepted.
- There will be no make-up quizzes—**three (3) worst quizzes will be dropped** before calculating the semester grade.
- **There will be 1 midterm exam, lasting 2 hours. The tentative date of the midterm is Saturday, Oct 25, from 9:00-11:00 am.** Closed-book, no cheat sheets.
- **Missing** the midterm or final exam without prior notification to and approval of me will automatically result in the **"0"** grade for the exam.

Course Contents:

Limit of a function and limit laws, one-sided limits, continuity, infinite limits, asymptotes of graphs
Tangents, rate of change and derivative at a point, derivative as a function, higher order derivatives, differentiation rules, derivative of trigonometric functions
Derivative of trigonometric functions(continued), chain rule, implicit differentiation, related rates
Linearization and differentials, extreme values of a function, mean value theorem, monotonicity, first derivative test
Concavity, second derivative test, curve sketching, applied optimization
Newton's method, area approximation by using left, right, and mid sums, limits of these sums, definite integral
Fundamental theorem of calculus, antiderivatives and indefinite integrals, substitution method
Substitution method (continued), area between curves, volumes using cross-sections, arclength, areas of surfaces of revolution
Work and fluid forces, inverse functions and their derivatives, natural log, exponential functions
L'Hôpital's rule, inverse trigonometric functions, relative rates of growth, integration by parts
Trigonometric integrals, trigonometric substitutions, partial fractions method
Partial fractions method (continued), numerical integration: by rectangle, trapezoidal and Simpson's rules, improper integrals
First order ODE: solutions, slope fields, Euler's method, linear equations, separable equations

First order ODE: applications--“mixing problems”, Malthusian model, Logistic model, autonomous equations, phase-line analysis

Homework Assignments:

- 2.1: #1, 7, 8, 15, 16
- 2.2: #1, 4, 9, 10, 18, 19, 23, 28, 32, 40, 44, 45, 54, 57, 65 (a), 75, 77, 80
- 2.3: skipped
- 2.4: #1, 5, 6, 17, 20, 28, 32, 34, 44, 45, 46
- 2.5: #7-10, 26, 27, 32, 38, 43, 47, 55, 65, 67, 70
- 2.6: #2, 10, 11, 16, 21, 23, 31, 36, 39, 43, 63, 67, 69, 70, 80, 82, 99

- 3.1: #1, 2, 23, 33, 41, 47
- 3.2: #1, 20, 27-31, 34, 43, 44, 46-48, 57, 58
- 3.3: #6, 11, 17, 20, 21, 29, 37, 38, 41, 43, 48, 55-58
- 3.4: #3, 7, 15, 21, 23, 25, 26
- 3.5: #3, 12, 15, 17, 26, 32, 34, 45, 50, 51, 56, 58, 59
- 3.6: #5, 7, 10, 15, 29, 34, 41, 54, 65, 70, 73, 74, 82, 84,
- 3.7: #1, 12, 19, 29, 41
- 3.8: #7, 23, 27, 32, 42
- 3.9: #1, 6, 17, 18, 29, 37, 42, 51, 56

- 4.1: #1-6, 20, 36, 42, 45, 50, 51, 67, 68
- 4.2: #13, 15 a, (iii), 15b, 18, 20, 23, 27, 33, 60, 61, 63, 65
- 4.3: #15-18, 23, 35, 55a, 65
- 4.4: #1, 4, 17, 23, 32, 46, 73, 76, 93, 95, 98, 99, 104
- 4.5: #1, 7, 12, 20a, 22, 28, 38, 48, 49, 51, 54, 67a
- 4.6: #1, 3, 7, 9, 10, 20
- 4.7: #1, 7, 12, 23, 32, 34, 52, 59, 63, 75, 85, 95, 99, 103

- 5.1: #1, 3, 5, 7, 9, 19
- 5.2: #8, 12, 30, 32, 33, 35, 37, 38, 39, 40
- 5.3: #1, 3, 6, 8, 9, 17, 20, 55, 71, 73, 76, 77, 78, 81, 83
- 5.4: #1, 3, 4, 8, 9, 10, 12, 17, 22, 24, 27, 28, 30, 40, 44, 46, 61, 63, 68, 71, 73
- 5.5: #18, 20, 21, 22, 26, 31, 38, 42, 43, 46, 56, 62
- 5.6: #8, 11, 12, 21, 23, 25, 59, 63, 71, 72, 78, 82, 83, 84-87

- 6.1: #1, 10, 12, 14, 19, 23, 27, 29, 37, 41, 43, 51, 59
- 6.3: #1, 2, 9, 23, 30
- 6.4: #13, 18, 22, 23, 25, 28
- 6.5: #2, 7, 10, 14, 23, 24, 29, 36, 44
- (6.2 and 6.6 skipped)

- 7.1: #1-6, 12, 14, 17, 25, 31, 38, 39, 44, 56, 59

- 7.2: #1, 4, 5, 8, 9, 21, 23, 34, 39, 42, 45, 53, 56, 65, 70, 73, 77, 85
- 7.3: #1, 8, 14, 16, 24, 26, 27, 29, 38, 48, 56, 59, 62, 65, 74, 75, 82, 85, 88,
91, 96, 99, 111, 118, 121, 123, 131, 136
- 7.4: delayed
- 7.5: #16, 20, 21, 26, 31, 38, 40, 41, 45, 51, 53, 59, 60, 61, 66, 67, 68, 75, 85, 88
- 7.8: (taught before 7.6) #1, 9-12, 20, 21(a), 22
- 7.6: #1, 3, 5, 7, 13, 16, 18, 22, 25, 33, 44, 47, 52, 63, 65, 69, 74, 87, 91, 92, 96, 97,
123, 127
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- 8.1: skipped
- 8.2: #2, 6, 12, 17, 23, 28, 29, 31, 32, 37, 43, 45, 50, 51, 53, 61, 63, 69
- 8.3: #3, 14, 15, 19, 20, 24, 27, 30, 33, 34, 35, 53, 54, 56, 64, 67
- 8.4: #2, 3, 5, 8, 9, 10, 15, 17, 21, 26, 36, 37, 57, 58
- 8.5: #6, 7, 13, 16, 20, 23, 27, 31, 34, 35, 41
- 8.7: #8, 24, 25, 28, 29, 31
- 8.8: #4, 10, 14, 20, 23, 27, 32, 35, 39, 41, 52, 53, 56, 61, 64, 65, 67, 72, 76
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- 9.1: #1-6, 11, 14 (do 11 and 14 after learning 7.4 and 9.2)
- 7.4: #11, 12, 13, 17, 30, 34, 42
- 9.2: #1, 4, 16, 17
- 9.3: #13, 15
- 9.4: #1, 7, 10, 12, 14, 18 (for these problems, no need to study the sign of the second derivative of the solutions)